

Lanpei Li

PhD Candidate in Computer Science • University of Pisa • HPC-Lab, ISTI-CNR

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RESEARCH INTERESTS

My doctoral research designs *adaptive, AI-based orchestration* that lets structured applications self-manage across the cloud-edge / IoT continuum, combining reinforcement learning, distributed systems and serverless / HPC platforms. Core topics include distributed resource management and resource management of the cloud continuum, reinforcement learning for autonomic computing and structured parallel systems, and foundation models for resource management, with a complementary line of work in continual / lifelong learning and continual parameter-efficient fine-tuning (PEFT).

EDUCATION

Ph.D. in Computer Science — University of Pisa

Nov 2023 – present (expected Oct 2026)

39th cycle • Scientific area INFO-01/C • research carried out at HPC-Lab, ISTI-CNR.

- **Curriculum / research line:** Distributed Management and Dependability of Structured Applications over the Cloud/IoT Continuum.
- **Supervisor:** Massimo Coppola (ISTI-CNR); **co-supervisor:** Vincenzo Lomonaco (LUISS University).
- **Official ISTI association:** May 2024 – present.
- **Research affiliations:** HPC-Lab, ISTI-CNR (Nov 2023–present) • ATLAS Lab, University of Pisa / LUISS University (Jan 2026–present).

M.Sc. in Computing Science (LM-18) — University of Pisa

Sep 2015 – Jul 2023

Final mark: 110/110.

- **Thesis:** *Continual Policy Distillation of RL-based Controllers for Soft Robotic In-Hand Manipulation* — advisors Vincenzo Lomonaco and Egidio Falotico (BRAIR Lab, The BioRobotics Institute, Scuola Superiore Sant'Anna).

B.Eng. in Software Engineering — Xi'an Institute of Posts & Telecommunications

Sep 2006 – Jul 2010

Final mark: 82/100.

PUBLICATIONS

1. L. Quarantiello, **L. Li**, E. Tavan, I. Testa, G. Carfì, G. Gramaglia, J. Bell, D. Malitesta, P. Averty, E. N. Coleman, V. Lomonaco. "Winter Is Over for Continual Learning: From Continual Representation Learning to Continual Learning Agents." *24th International Conference of the Italian Association for Artificial Intelligence (AixIA), discussion paper track*, Perugia, Italy, 2026. CEUR-WS (to appear); preprint [doi:10.20944/preprints202607.1116.v1](https://doi.org/10.20944/preprints202607.1116.v1)
2. **L. Li**, A. Vaccarella, V. Lomonaco, M. Coppola. "ContinuumBench: Benchmarking Joint Autoscaling and Placement Across Evaluation Regimes in the Cloud-Edge Continuum." *6th Workshop on Flexible Resource and Application Management on the Edge (FRAME), co-located with the 32nd International European Conference on Parallel and Distributed Computing (Euro-Par)*, Pisa, Italy, 2026.
3. A. Vaccarella, **L. Li**, V. Lomonaco, M. Coppola. "Smart Adaptive Computing Across the Continuum: LLMs in IoT-Edge-Cloud Resource Management." *6th Workshop on Flexible Resource and Application Management on the Edge (FRAME), co-located with Euro-Par*, Pisa, Italy, 2026.
4. G. Di Palma*, A. Pardini*, **L. Li***, G. Pompei*, M. Coppola, V. De Caro, V. Lomonaco. "On-Board Efficiency: Comparing Compressed Deep Learning and Classical Models for Earth Observation." *Pattern Analysis and Applications* 29(4), 185, Springer, 2026. [doi:10.1007/s10044-026-01720-0](https://doi.org/10.1007/s10044-026-01720-0) (*joint first authors)
5. **L. Li**, M. Coppola, M. Li, V. Besozzi, J. Bell, V. Lomonaco. "Reinforcement Learning-Based Dynamic Management of Structured Parallel Farm Skeletons on Serverless Platforms." *34th Euromicro International Conference on Parallel, Distributed and Network-based Processing (PDP)*, 2026.
6. **L. Li**, J. Bell, M. Coppola, V. Lomonaco. "Adaptive AI-based Decentralized Resource Management in the Cloud-Edge Continuum." *33rd Euromicro International Conference on Parallel, Distributed and Network-based Processing (PDP)*, Turin, Italy, 2025, pp. 329–332. [doi:10.1109/PDP66500.2025.00053](https://doi.org/10.1109/PDP66500.2025.00053)
7. J. Bell, L. Quarantiello, E. N. Coleman, **L. Li**, M. Li, M. Madeddu, E. Piccoli, V. Lomonaco. "The Future of Continual Learning in the Era of Foundation Models: Three Key Directions." *Workshops at the 4th International Conference on Hybrid Human-Artificial Intelligence (HHAI)*, CEUR-WS Vol-4074, 2025, pp. 525–548.

8. L. Quarantiello, E. Piccoli, J. Bell, M. Li, G. Carfì, E. N. Coleman, G. Gramaglia, **L. Li**, M. Madeddu, I. Testa, V. Lomonaco. "A Compositional Paradigm for Foundation Models: Towards Smarter Robotic Agents." *7th Italian Conference on Robotics and Intelligent Machines (I-RIM)*, 2025, pp. 255–257. doi:10.5281/zenodo.17629900
9. **L. Li**, E. Piccoli, A. Cossu, D. Bacciu, V. Lomonaco. "Calibration of Continual Learning Models." *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, Seattle, WA, USA, 2024, pp. 4160–4169. doi:10.1109/CVPRW63382.2024.00419
10. **L. Li**, E. Donato, V. Lomonaco, E. Falotico. "Continual Policy Distillation of Reinforcement Learning-based Controllers for Soft Robotic In-Hand Manipulation." *IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 1026–1033. doi:10.1109/RoboSoft60065.2024.10522027

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Researcher — University of Pisa / ISTI-CNR

Nov 2023 – present

- **HPC-Lab, ISTI-CNR** (Nov 2023–present):
 - Research on cloud-edge / cloud-continuum resource management, serverless orchestration and structured parallel skeletons; published work at PDP 2025, PDP 2026 and FRAME @ Euro-Par 2026.
 - Designed benchmarking methodology for cloud-continuum resource management, focusing on reproducible evaluation of orchestration policies across heterogeneous edge, cloud and serverless infrastructures.
 - Investigating LLM-assisted management of cloud-continuum systems, including how language models can support workload characterization, deployment decisions and adaptive resource-control workflows.
 - Developing foundation-model-based controllers for resource management, exploring how pretrained models can be adapted to scheduling, scaling and orchestration tasks in dynamic distributed environments.
- **ATLAS Lab (formerly COLLAGE Lab), University of Pisa / LUISS University** (Nov 2025–present):
 - Research on continual / lifelong learning and efficient adaptation of foundation models, with emphasis on parameter-efficient fine-tuning under sequential data and task shifts.
 - Studying stability and forgetting in continual adaptation, including methods for maintaining model reliability when updating foundation models over time without full retraining.

Research Intern — China Telecom Beijing Research Institute

Sep 2020 – Mar 2023

- Supported research on the **Computing Power Network** (compute–network convergence) through technical literature review and internal summaries.

Camera Module Dev. Engineer & Project Manager — Lenovo (Beijing) Co., Ltd.

Oct 2011 – Jan 2015

- Developed and maintained camera HAL/driver layers (Intel platform, Android); led feature development, image-quality tuning and project planning.

BSP Development Engineer — MobileXP Technology Co., Ltd.

Aug 2010 – Aug 2011

- Developed device drivers (keypad, camera, battery/charger) for mobile devices on the Qualcomm/Android platform; bring-up of new hardware build environments.

MENTORING & SUPERVISION

- **B.Sc. thesis co-supervision**, University of Pisa, Mar–Jul 2024 — *Machine Learning Model Compression for On-Board Earth Observation* (student: G. Di Palma); work extended into a co-authored article in *Pattern Analysis and Applications*, 2026.

TALKS & POSTERS

- **Talk**: "Adaptive AI-based Decentralized Resource Management in the Cloud–Edge Continuum," AHPC³ Workshop @ Euromicro PDP 2025, Turin, Italy, Mar 2025.
- **Talk**: "RL-based Dynamic Management of Structured Parallel Farm Skeletons on Serverless Platforms," AHPC³ Workshop @ Euromicro PDP 2026.
- **Talk**: "ContinuumBench: Benchmarking Joint Autoscaling and Placement Across Evaluation Regimes in the Cloud–Edge Continuum," FRAME Workshop @ Euro-Par 2026, Pisa, Italy, Aug 2026.
- **Poster**: PhD Spotlight Event, University of Pisa, Nov 2025.

ACADEMIC SERVICE

Peer reviewer: ACM HPDC 2026 • IEEE ICRA 2026 • IEEE RoboSoft 2026 • CoLLAs 2026 (Conference on Lifelong Learning Agents) • FRAME 2024 Workshop @ ACM HPDC 2024.

Conference organization (student volunteer): ACM HPDC 2024, Pisa.

TRAINING

Summer schools: ACM Europe Summer School on HPC — Computer Architectures for AI & Dedicated Applications, Barcelona Supercomputing Center, Spain, Sep 2024 (*Certificate of Honor for Outstanding Performance*); Lipari School on Computational Complex & Social Systems, Lipari, 2024; 2nd CINI National PhD School on HPC & Emerging Technologies, University of Trento, 2024.

PhD courses: Cloud Computing & Big Data (Scuola Superiore Sant'Anna); Programming Tools & Techniques in the Pervasive Parallelism Era; Big Data Analytics; Pathway to Green ICT; Data Compression & Compressed Data Structures; Collective Machine Intelligence.

Seminars & doctoral training: English for Research Publication and Presentation Purposes; English for Scientific Communication using AI; Writing and Defending the PhD Dissertation in the Age of AI: An Interdisciplinary Perspective.

TECHNICAL SKILLS

Programming: Python, C/C++.

Distributed & parallel systems: structured parallel skeletons, serverless platforms, Linux systems & HPC cluster environments.

Machine learning: Reinforcement learning; Continual / Lifelong learning; Parameter-Efficient Fine-Tuning (PEFT); Foundation models; Computer Vision; LLMs.

Systems / embedded: Linux kernel and device-driver development (Android camera architecture).

LANGUAGES

Chinese (native) • English (fluent)